

1. Abstract

Local events play an essential role in community life. They **connect people, foster relationships, and encourage cultural, educational, and professional growth** within neighborhoods and institutions. However, many small community events go unnoticed, as traditional promotion methods (flyers, sporadic social media posts) are inconsistent and hard to track. Major online platforms like Eventbrite emphasize large-scale or commercial events, leaving smaller local gatherings underrepresented. In fact, one industry analysis explicitly notes that *“people miss local events because discovery is fragmented across platforms,”* highlighting a need for a centralized solution. In response to this gap, we propose a **mobile-first app for local event discovery**. This app will aggregate events from nearby organizers and allow users to **search, filter, and book** community happenings through an intuitive interface, thereby bridging the divide left by existing platforms. The goal is to create a one-stop hub where both attendees and event organizers can easily connect, improving visibility for local events and strengthening community engagement.

2. Problem Statement

Despite the abundance of local activities (concerts, workshops, sports, festivals), **current discovery methods are fragmented and inefficient**. Popular services (Eventbrite, Meetup, etc.) tend to highlight large or paid events, while neighborhood gatherings rely on scattered flyers or social media. This means **community events are often overlooked by potential attendees**. The fragmentation forces event organizers to use multiple channels without real-time visibility, leading to lower attendance and reduced engagement. Furthermore, **existing apps may incur high costs or technical barriers**; for example, Google Maps API fees can be significant for mapping features, and many event platforms are not tailored for local, grassroots events. In summary, the problem is: **How can we provide a cost-effective, unified platform that makes local events easy to discover and manage?** We address this by focusing on the local community: offering free map integration (via OpenStreetMap), user-friendly filters, and dedicated organizer tools. By solving the fragmentation (as noted in industry research), we aim to increase event visibility and attendance in our area.

3. Objectives of the Project

The primary objectives of the Local Events app are:

- **Centralized Event Listings:** Create a platform where all local events (free or paid) within a region can be listed and browsed.

- **Powerful Search and Filters:** Implement comprehensive search by keywords, and filters by category, date, location radius, price (free/paid), and popularity to help users quickly find relevant events.
- **Rich Event Details:** Provide detailed event pages with descriptions, images, organizer info, seating or ticket availability, and contact details.
- **Secure Ticket Booking:** Integrate reliable payment gateways (Razorpay for India, Stripe for global) to allow in-app ticket purchases, ensuring smooth transactions.
- **Interactive Maps Integration:** Display event locations on interactive maps (using OpenStreetMap tiles), enabling users to see and navigate to nearby events.
- **User Authentication & Profiles:** Implement secure user login (JWT tokens) and social sign-in (Google Sign-In) so attendees and organizers have personalized accounts.
- **Organizer Dashboard:** Provide a web interface for event creators to manage their events, upload media, and track registrations.
- **Admin Moderation and Analytics:** Include tools for app administrators to review events, prevent fraud, and view usage analytics.
- **Real-time Notifications:** Send push notifications for upcoming events, ticket confirmations, and important updates to keep users engaged.
- **Scalability and Cost Efficiency:** Use a free-tier friendly tech stack (e.g. MongoDB Atlas, Render, Netlify, Cloudinary) and cross-platform code reuse to minimize hosting costs and development time.

These objectives ensure the app not only lists events but also actively **facilitates event management and community engagement**. By meeting these goals, the app will streamline the entire event lifecycle—from discovery to registration—for both users and organizers.

4. Scope of the Project

The scope defines what the app will cover and what it will not. This project will focus on **mobile (Android and iOS) and web platforms** for local event discovery, with the following boundaries:

- **Inclusions:**
 - **Geographic Scope:** Local community or city-level events. Initially, the app will focus on a specific region or campus, expanding to additional areas over time.

- **User Roles:** Three main roles: Attendees (who search for and attend events), Organizers (who create and manage events), and Admin (who oversees content and usage).
- **Functionalities:** Event listing and search, filtering, detailed event pages, booking tickets, payment processing, map-based discovery, user authentication, notification system, and basic analytics.
- **Platforms:** Cross-platform mobile app (built with React Native/Expo) and a web-based admin/organizer dashboard (built with React).
- **Access Requirements:** The app requires internet access and device permissions for location and notifications.
- **Exclusions:**
 - **Global/International Events:** The system will not focus on international ticketing or large-scale global events; it targets local/community gatherings.
 - **Physical Ticket Printers:** It assumes fully digital ticketing; printing hardware integration is outside scope.
 - **Augmented Reality or VR:** The app will not use AR/VR technologies for event visualization.
 - **Payment Processors beyond Razorpay/Stripe:** Other payment methods (like PayPal) are not prioritized.
 - **Massive Scalability & Big Data:** While designed to scale, features like complex event recommendation engines or AI-driven personalization are beyond the initial scope.

By clearly defining scope, we ensure resources are focused on building the core local-event discovery functionality. Future work (outside current scope) could include expanding to larger regions, advanced personalization, or integrating new payment options, but the initial version will deliver a robust platform for local communities.